

Grain Growth with Active Parameter Tracking (APT)

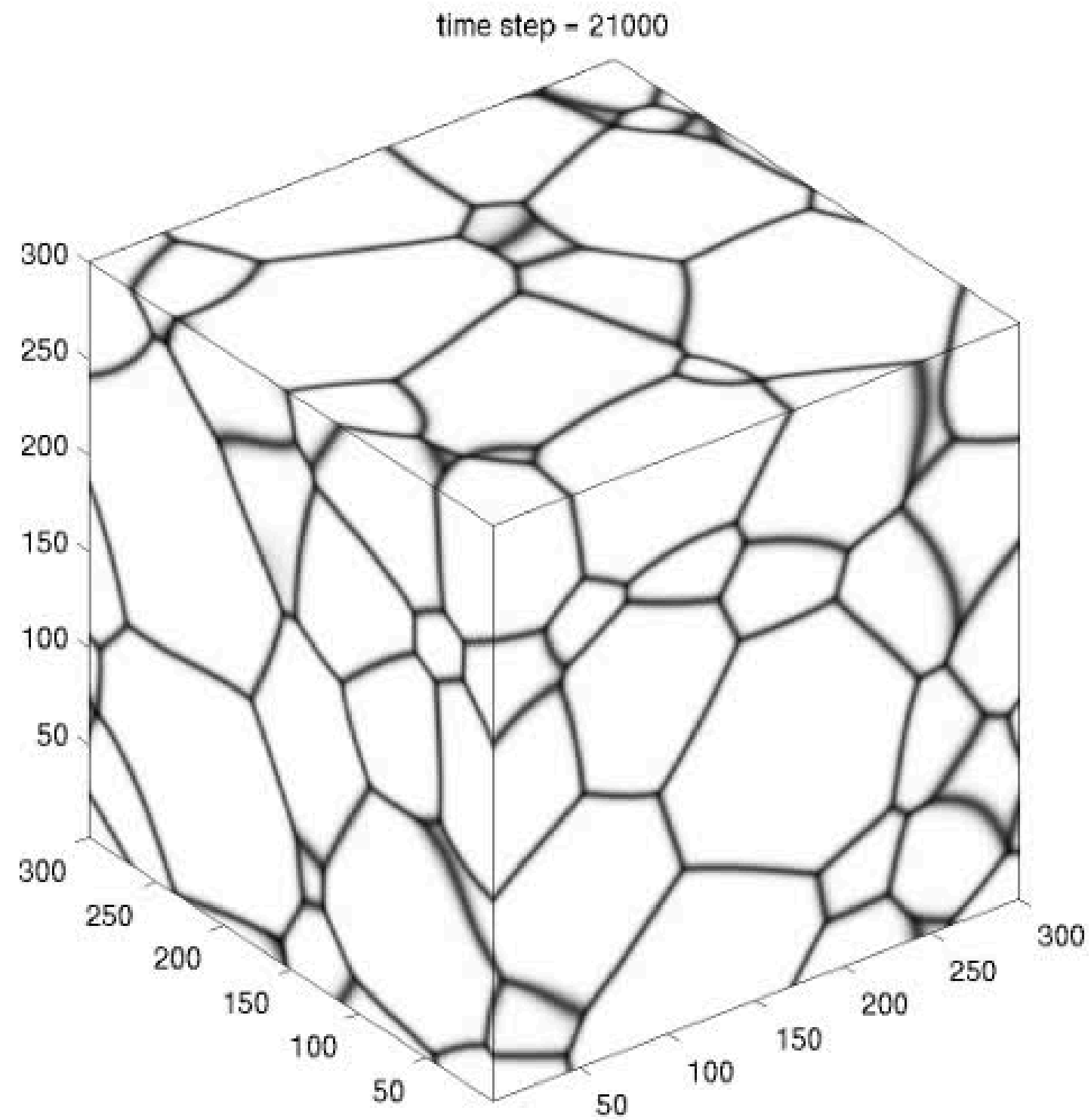
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Group Members:

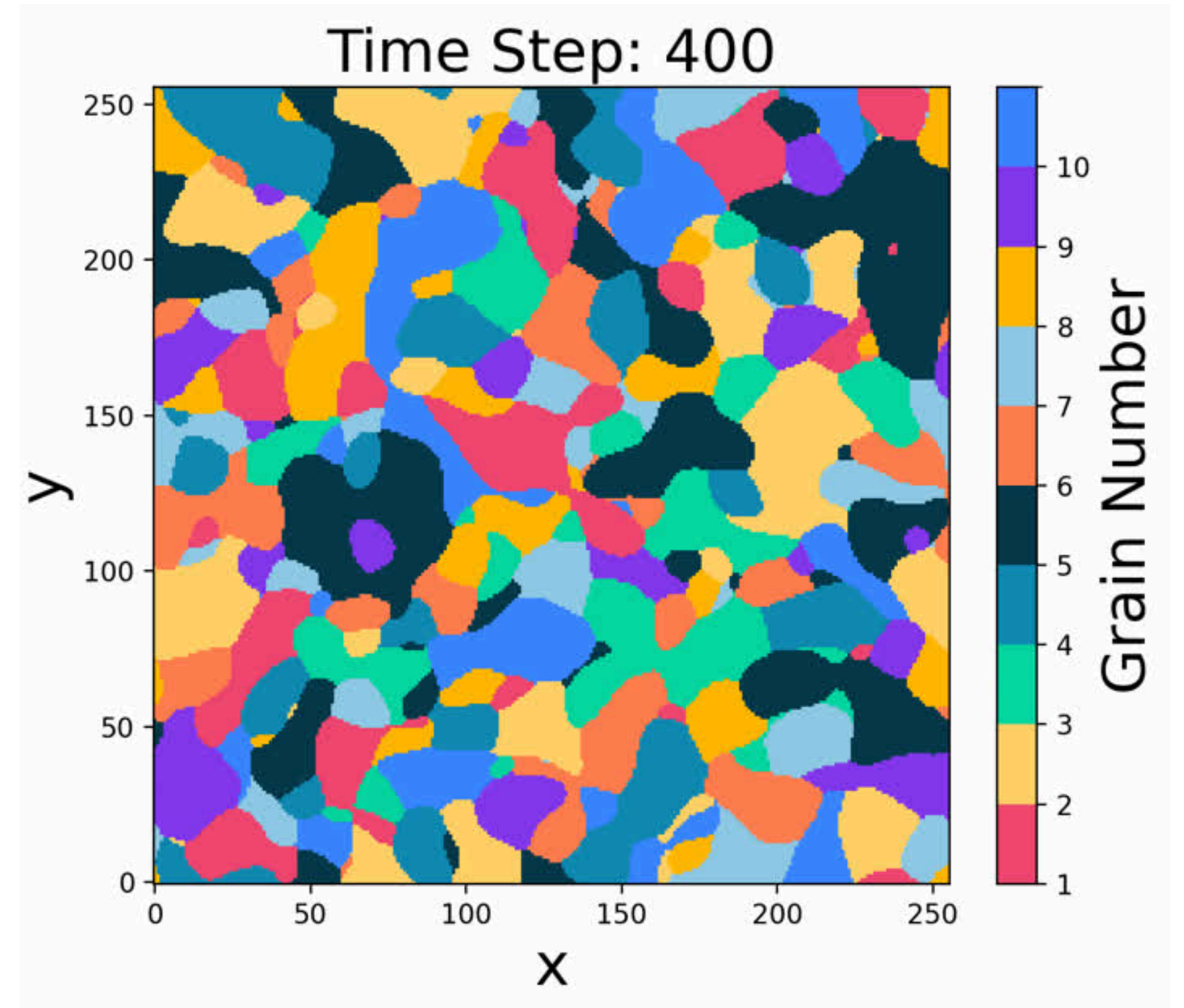
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Ashish Bhardwaj

Grain Growth



Courtesy: [/wiki/Grain_growth](https://en.wikipedia.org/wiki/Grain_growth)



Outline

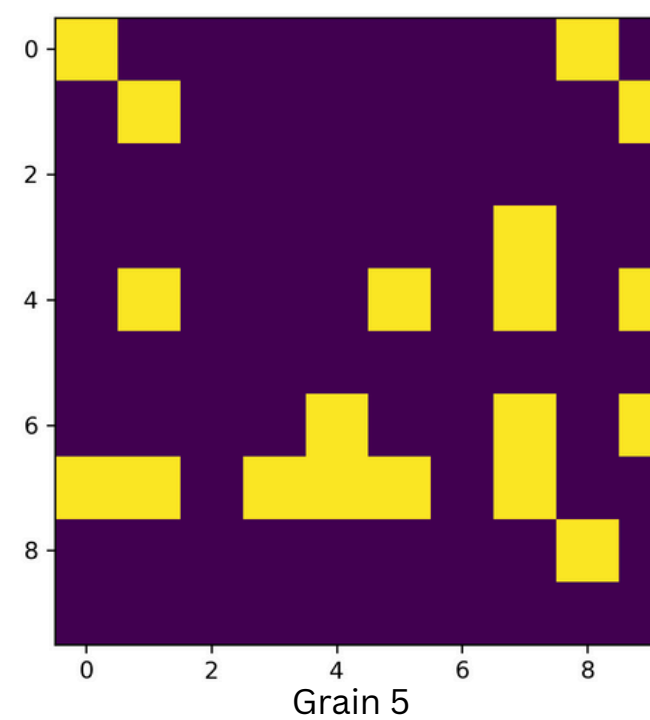
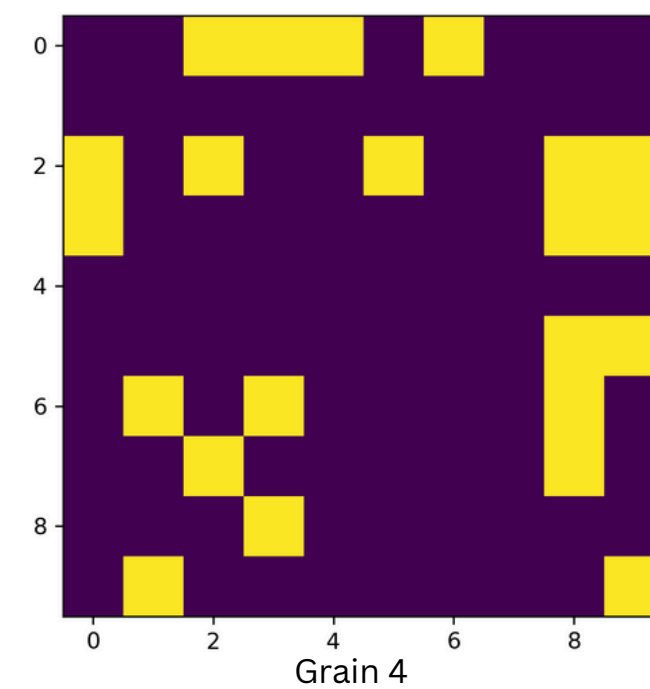
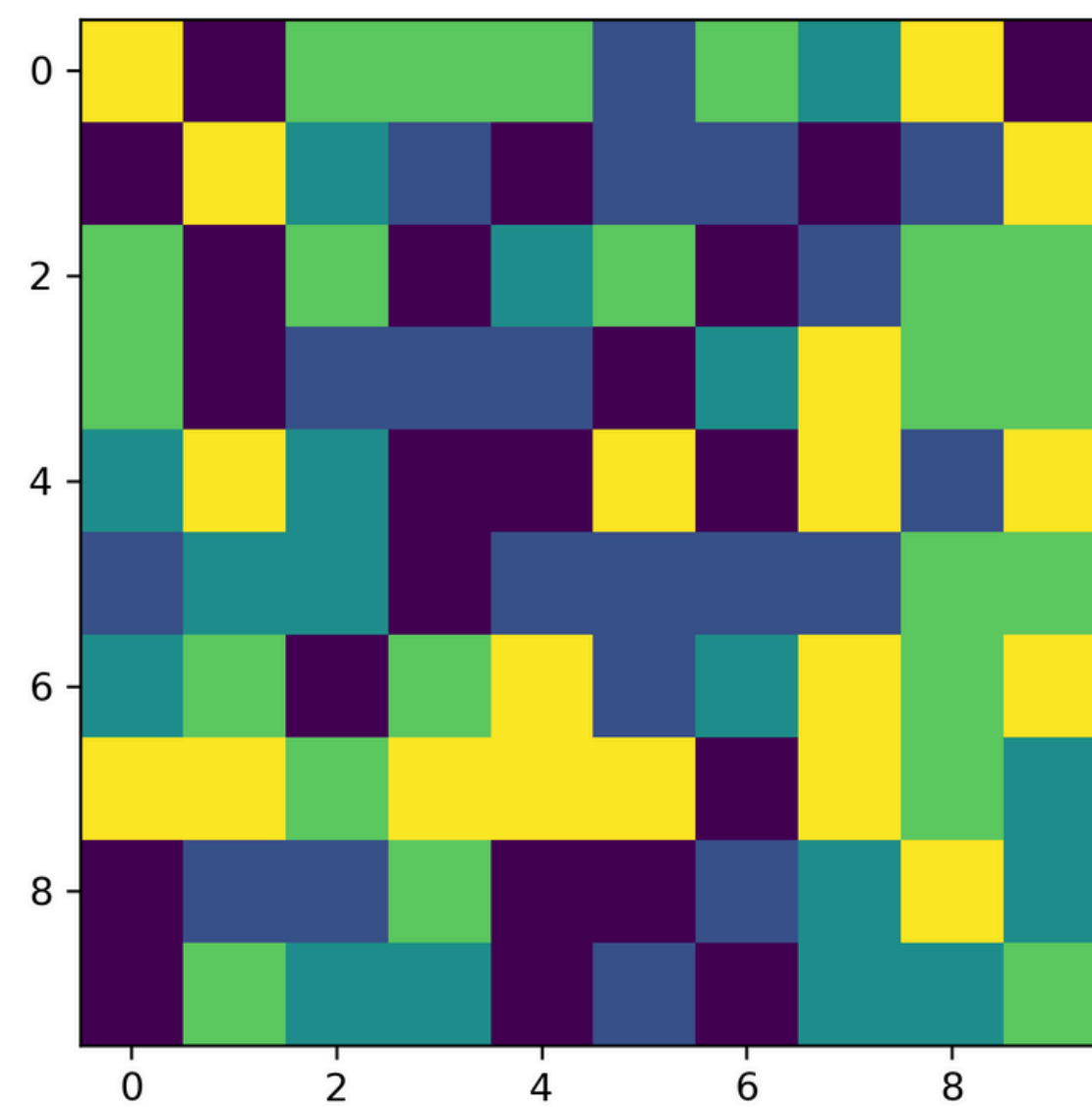
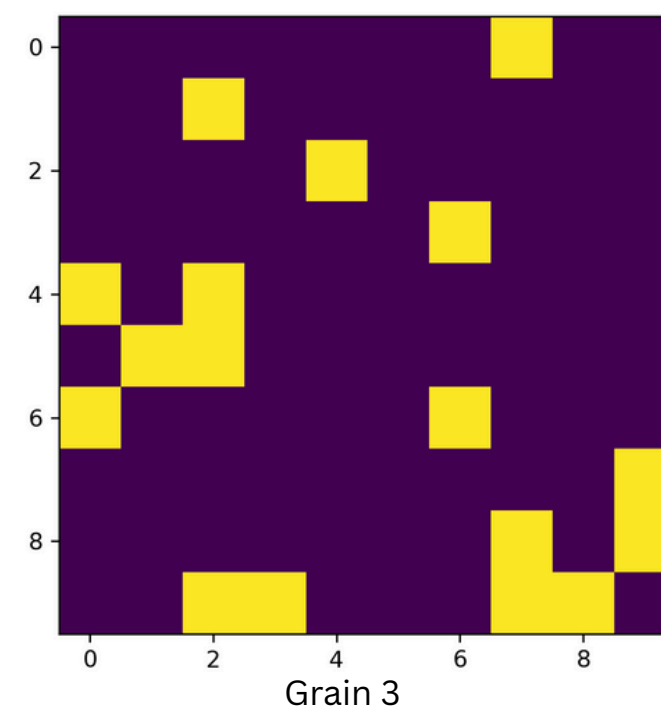
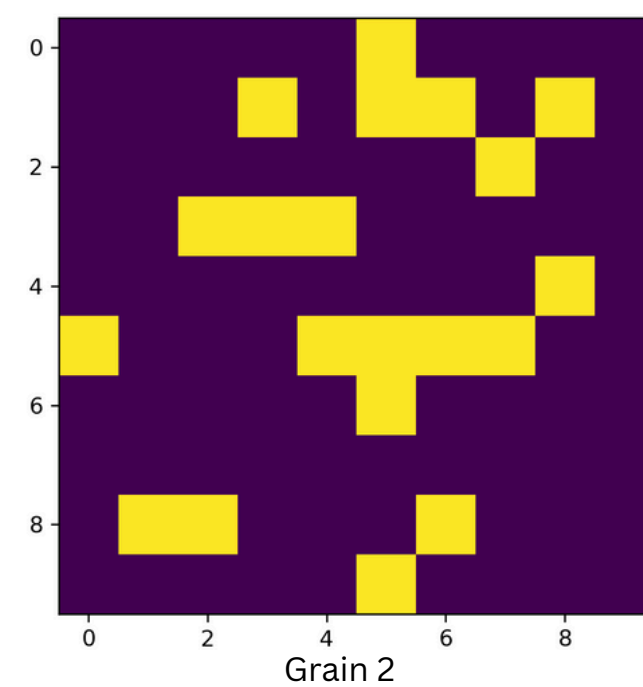
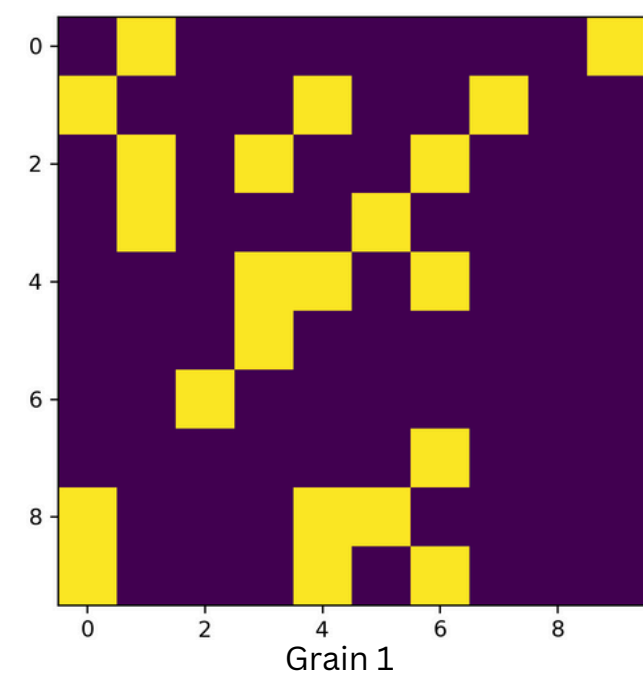
Grain Growth using:

- Finite Difference Method
- Fourier Spectral Method
- Active Parameter Tracking

$$\frac{\partial \eta_i}{\partial t} = -\mathcal{L}_i \cdot \frac{\delta F}{\delta \eta_i} \quad i = 1, 2, \dots, N$$

Governing Equation: Multivariate Allen-Cahn Equation

Initialization



Fourier Spectral Method

for each time step **do**

Initialize all $\eta_g^{\text{new}} \leftarrow 0$

for each grain $g = 1$ to N_{grains} **do**

Compute sum of squares of other grains:

$$S = \sum_{\substack{p=1 \\ p \neq g}}^{N_{\text{grains}}} \eta_p^2$$

Compute driving force term:

$$G = -\alpha\eta_g + \beta\eta_g^3 + 2\gamma\eta_g \cdot S$$

Take FFT of G and η_g :

$$\hat{G} = \mathcal{F}(G), \quad \hat{\eta}_g = \mathcal{F}(\eta_g)$$

Update in Fourier space:

$$\hat{\eta}_g^{\text{new}} = \frac{\hat{\eta}_g - \Delta t \cdot L \cdot \hat{G}}{1 + \Delta t \cdot L \cdot \kappa \cdot |\vec{k}|^2}$$

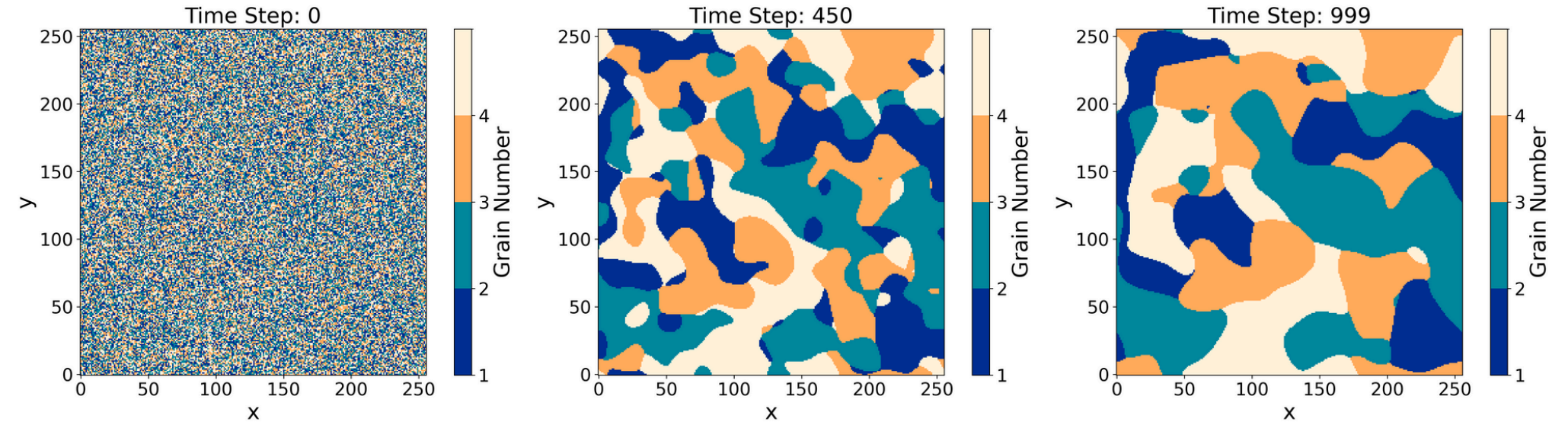
Transform back to real space:

$$\eta_g^{\text{new}} = \mathcal{F}^{-1}(\hat{\eta}_g^{\text{new}})$$

end for

end for=0

$$\frac{\partial \eta_i}{\partial t} = -\mathcal{L}_i \left(-\alpha\eta_i + \beta\eta_i^3 + 2\gamma\eta_i \sum_{j \neq i}^N \eta_j^2 - \kappa_i \nabla^2 \eta_i \right) \quad i = 1, 2, \dots, N$$



```
for step in range(nstep):
    etas_new = np.zeros((Ngrains, Nx, Ny))

    for i in range(1, Ngrains + 1):
        eta_i = eta[i]
        sum_eta_j_sq = np.sum([eta[j]**2 for j in range(1, Ngrains + 1) if j != i], axis=0)

        g = -alpha * eta_i + beta * eta_i**3 + 2 * gamma * eta_i * sum_eta_j_sq
        g_hat = np.fft.fft2(g)
        eta_hat = np.fft.fft2(eta_i)

        numerator = eta_hat - dt * L * g_hat
        denominator = 1 + dt * L * kappa * kpow2
        eta_i_new = numerator / denominator
        etas_new[i-1] = np.real(np.fft.ifft2(eta_i_new))
```

Finite Difference Method

$$\frac{\partial \eta_i}{\partial t} = -\mathcal{L}_i \left(-\alpha \eta_i + \beta \eta_i^3 + 2\gamma \eta_i \sum_{j \neq i}^N \eta_j^2 - \kappa_i \nabla^2 \eta_i \right) \quad i = 1, 2, \dots, N$$

for each time step **do**

Initialize all $\eta_g^{\text{new}} \leftarrow 0$

for each grain $g = 1$ to N_{grains} **do**

Compute sum of squares of other grains:

$$S = \sum_{\substack{p=1 \\ p \neq g}}^{N_{\text{grains}}} \eta_p^2$$

Compute driving force term:

$$G = -\alpha \eta_g + \beta \eta_g^3 + 2\gamma \eta_g \cdot S$$

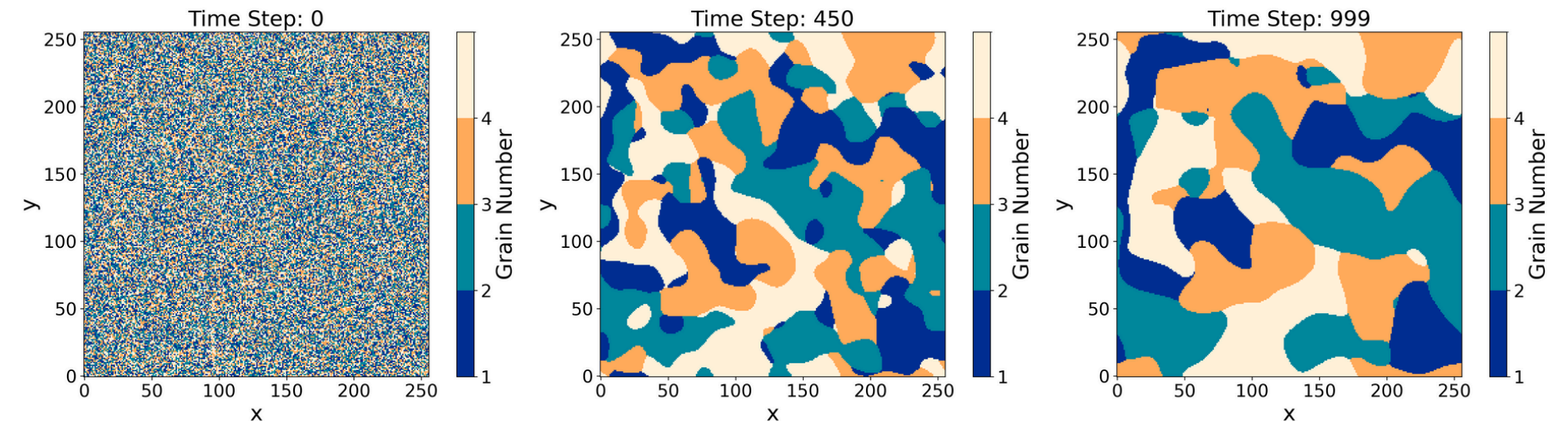
Compute Laplacian: $\nabla^2 \eta_g$

Update field:

$$\eta_g^{\text{new}} = \eta_g - \Delta t \cdot L \cdot (G - \kappa \nabla^2 \eta_g)$$

end for

end for=0



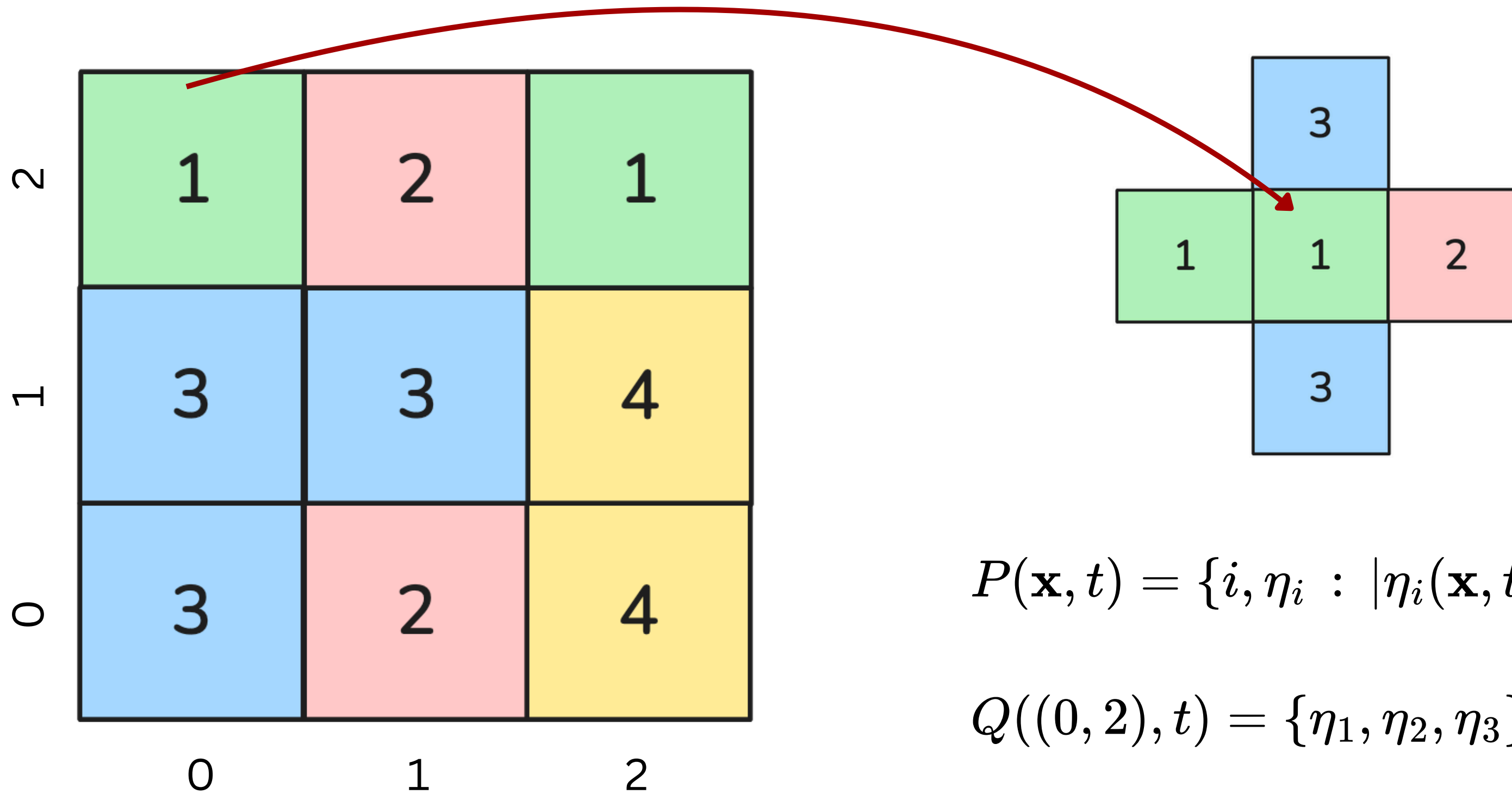
```
for step in range(nstep):
    etas_new = np.zeros((Ngrains, Nx, Ny))

    for i in range(1, Ngrains + 1):
        eta_i = eta[i]
        sum_eta_j_sq = np.sum([eta[j]**2 for j in range(1, Ngrains + 1) if j != i], axis=0)

        g = -alpha * eta_i + beta * eta_i**3 + 2 * gamma * eta_i * sum_eta_j_sq
        lap_eta = laplacian(eta_i, dx, dy)

        eta_i_new = eta_i - dt * L * (g - kappa * lap_eta)
        etas_new[i-1] = eta_i_new
```

Active Parameter Tracking (APT)



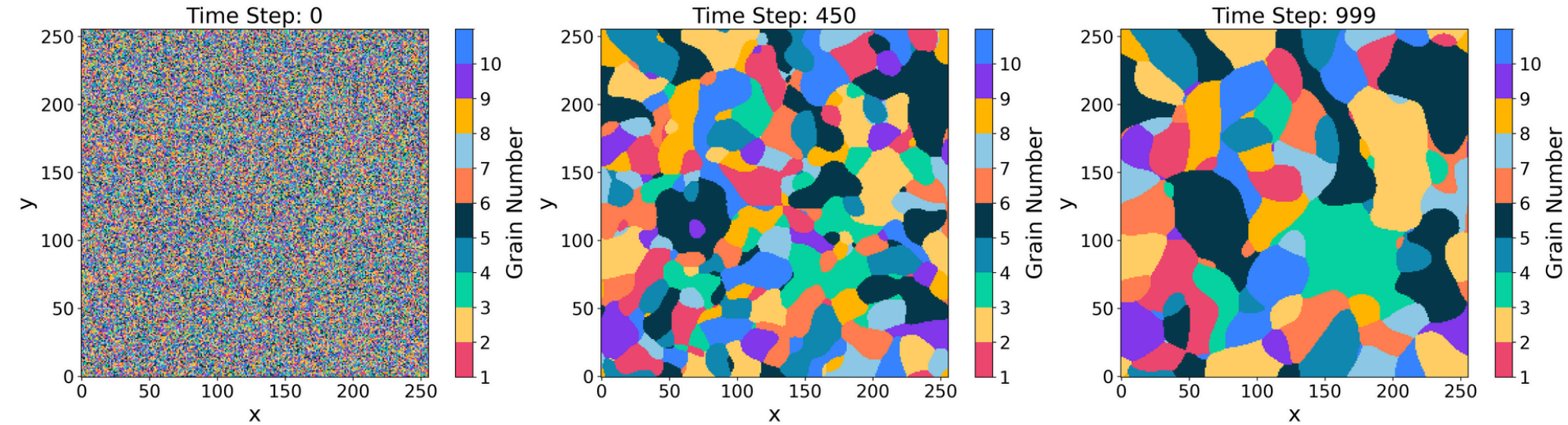
Number of Calculations using FDM = $N_x \times N_y \times N_{grains} \times N_{timestep}$

Number of Calculations using APT = $N_x \times N_y \times n \times N_{timestep}$

where $n \leq N_{grain}$

- n is the average number of active grains in the sample at a particular time step.
- At $t = 0$, $n \sim N_{grains}$. As the system evolves, n goes down due to grain growth

Active Parameter Tracking (APT)



for each time step **do**

Initialize all $\eta_g^{\text{new}}(i, j) \leftarrow 0$

for each grid point (i, j) **do**

$P(i, j) \leftarrow \{g \mid |\eta_g(i, j)| > \epsilon\}$

$Q(i, j) \leftarrow$ union of P values in 3×3 neighborhood of (i, j)

for each grain $g \in Q(i, j)$ **do**

Compute $g_{\text{term}} \leftarrow -\alpha\eta_g + \beta\eta_g^3 + 2\gamma\eta_g \sum_{\substack{p \in Q(i, j) \\ p \neq g}} \eta_p^2$

Compute $\nabla^2 \eta_g$ using finite difference

Update:

$$\eta_g^{\text{new}}(i, j) \leftarrow \eta_g(i, j) - \Delta t \cdot L \cdot (g_{\text{term}}$$

end for

for each grain $g \notin Q(i, j)$ **do**

$\eta_g^{\text{new}}(i, j) \leftarrow \eta_g(i, j)$

end for

end for

end for=0

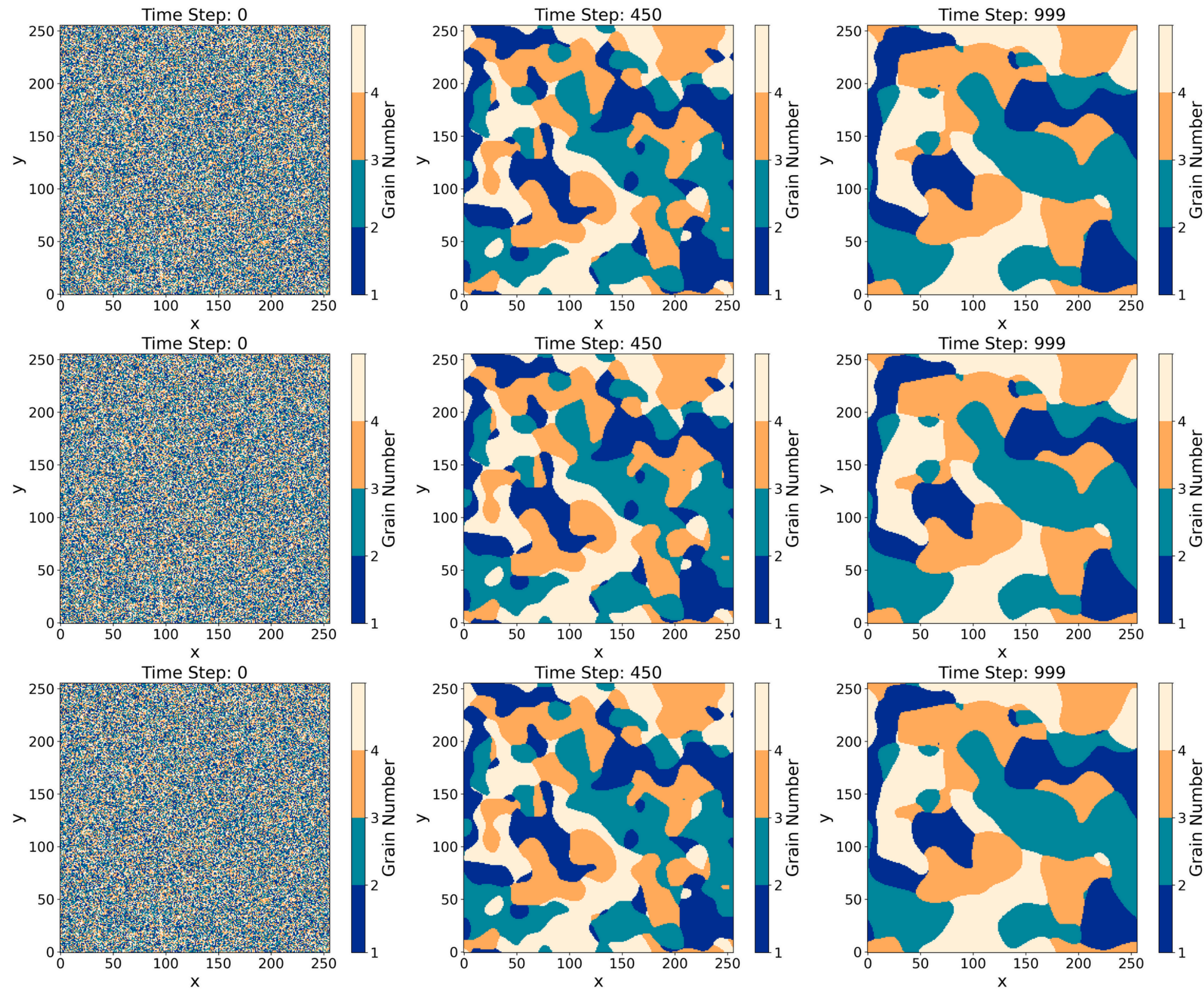
```
for step in range(nstep):
    etas_new = np.zeros((Ngrains, Nx, Ny))

    for i in range(Nx):
        for j in range(Ny):

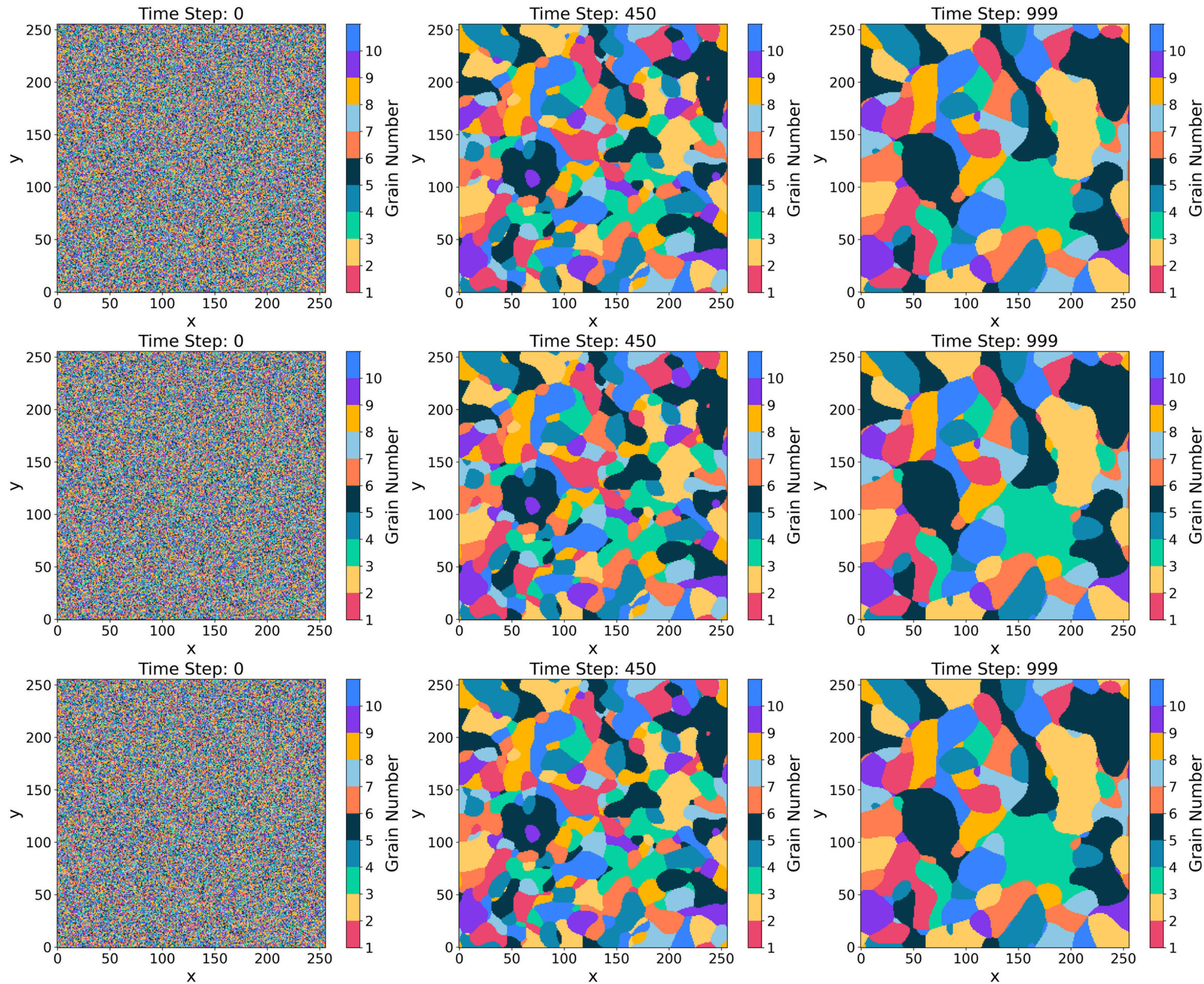
            # set P(x,t) at (i,j)
            P = {g for g in range(1, Ngrains + 1) if abs(eta[g][i, j]) > epsilon}

            # Q(x,t) from neighbors
            Q = set()
            for dx_, dy_ in neighbor_offsets:
                ni, nj = i + dx_, j + dy_
                if 0 <= ni < Nx and 0 <= nj < Ny:
                    for g in range(1, Ngrains + 1):
                        if abs(eta[g][ni, nj]) > epsilon:
                            Q.add(g)
```

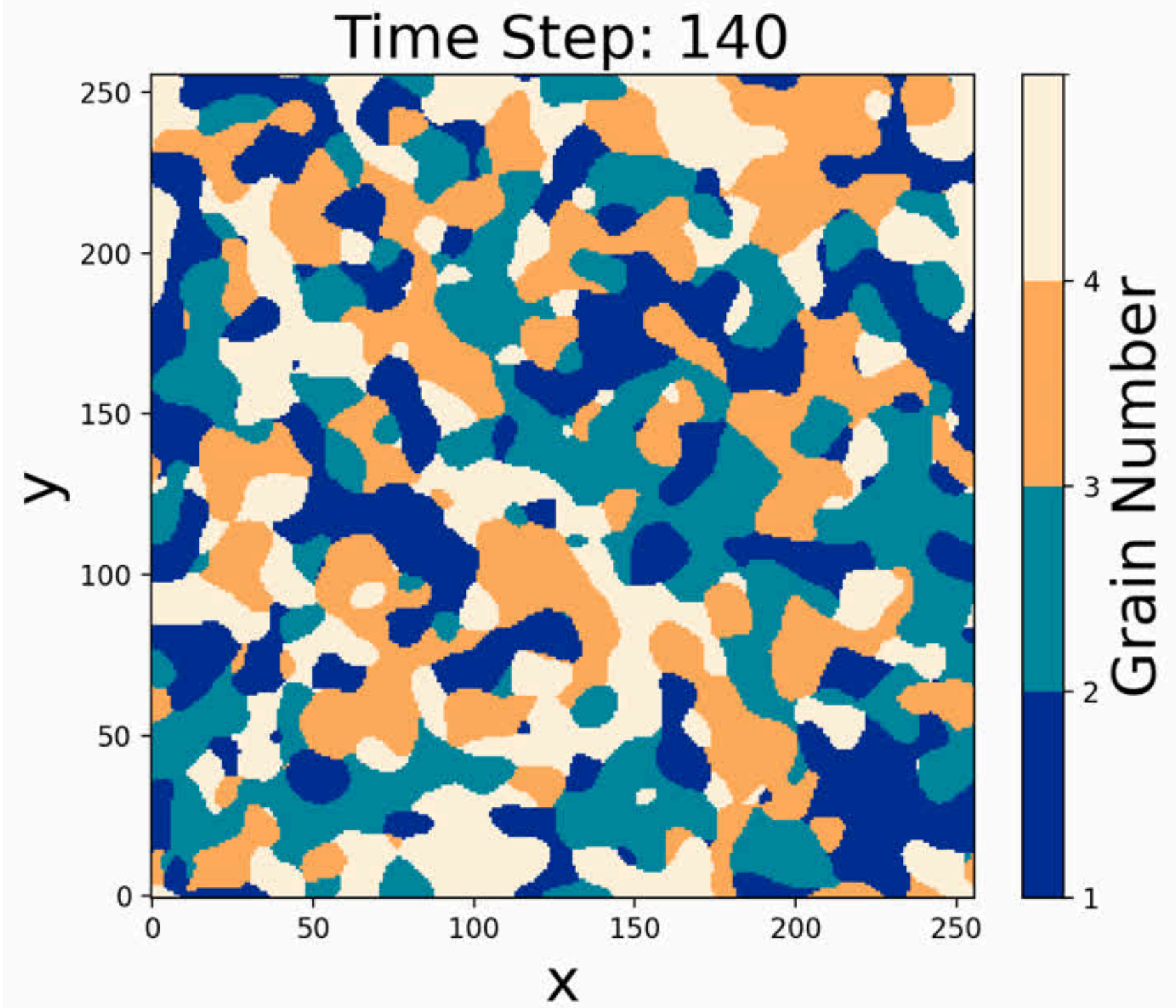

Ngrain = 4



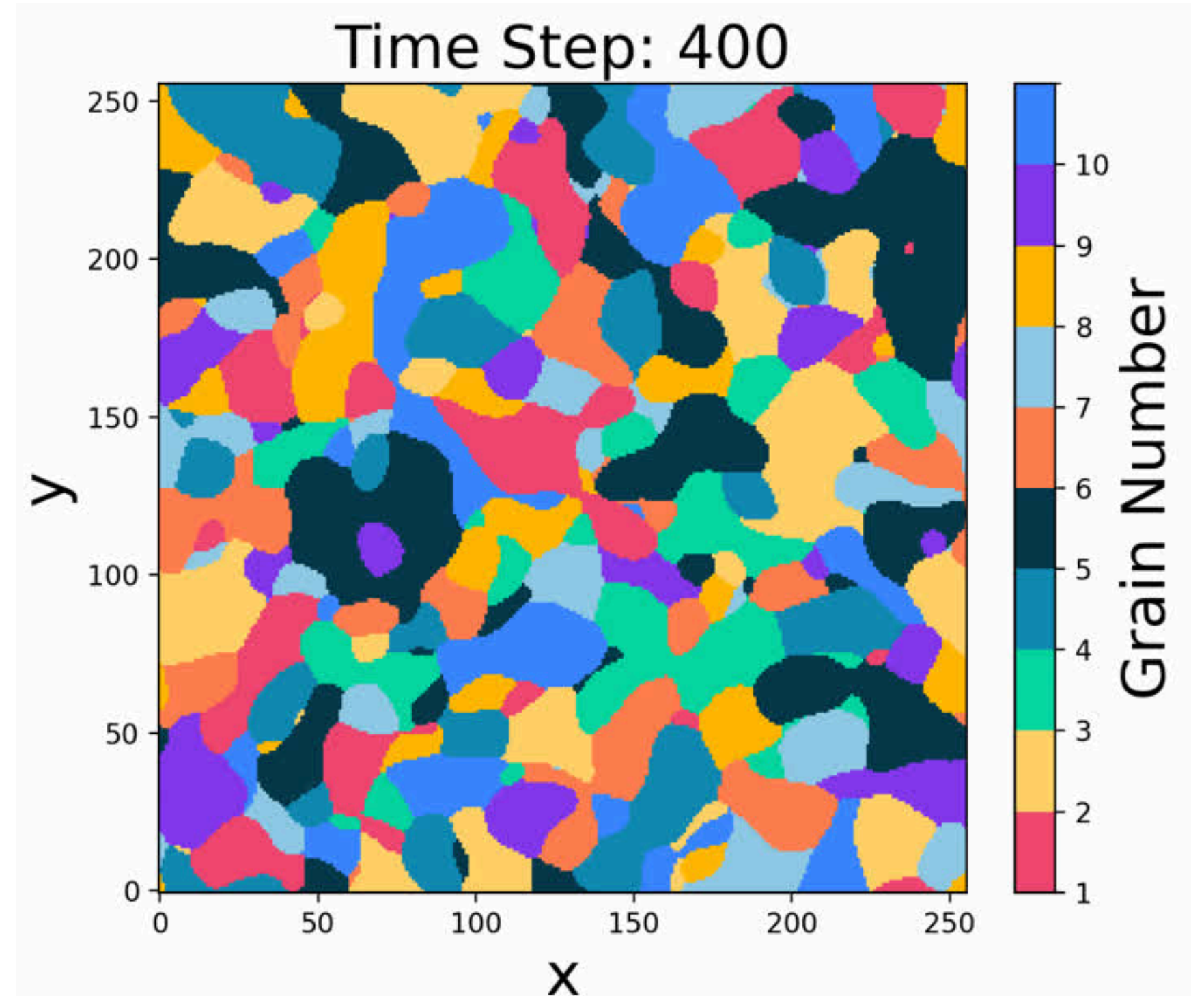
Ngrain = 10



Ngrain = 4



Ngrain = 10



Thank You

Find the presentation link [here](#)